

Autonomous Disaster Response Planner

Team: IBM Vanguard

Track: Open Innovation / AI

An AI-powered incident management system utilizing IBM watsonx and Google OR-Tools to optimize disaster resource allocation.

1. The Problem

Chaos in Crises: During natural disasters, information flows in raw, unstructured formats (radio transcripts, frantic texts, social media).

Inefficient Deployment: Dispatchers struggle to parse requests manually, leading to overlapping assignments or unfulfilled critical needs.

Why it matters: In emergency response, seconds lost in logistics cost lives.

2. Our Solution

Intelligent Ingestion: Uses IBM watsonx.ai Granite 3.0 to automatically extract structured data (urgency, location, need type) from messy field reports.

Mathematical Optimization: Employs a Constraint Programming solver (OR-Tools) to mathematically guarantee the optimal deployment of ambulances, trucks, and rescue boats.

Concurrency Control: Utilizes PostgreSQL row-level locking to ensure atomic dispatches, preventing double-booking even under massive load.

3. Architecture

- Frontend: React + Vite (TypeScript) for the interactive dispatcher dashboard.
- Backend: Python FastAPI for high-performance API routing.
- AI Engine: ibm-watson-machine-learning SDK hooked to Granite 3.0 models.
- Optimizer: Google OR-Tools CP-SAT Solver.
- Database: PostgreSQL with PostGIS for geospatial zone routing.
- Orchestration: Docker Compose.

4. Key IBM Technology Used

IBM watsonx.ai: Leveraged the ibm-watson-machine-learning SDK to parse raw incoming field reports.

By prompting the Granite 3.0 model, we bridged the gap between unstructured emergency chatter and strictly typed database records (Needs).

5. Results & Impact

Speed: Reduces dispatcher cognitive load and time-to-deployment from minutes to milliseconds.

Efficiency: The OR-Tools allocator prevents dispatching a small rescue boat to a massive flood zone if a heavy boat is available.

Scalability: Dockerized and ready to deploy to IBM Cloud or AWS.